Step 1 — Import libraries

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LogisticRegression
```

Step 2 — Create a simple DataFrame

```
data = {
    "Study_Hours": [1, 2, 2, 3, 3, 4, 4, 5, 6, 7, 8,
9],
    "Pass":        [0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1]
}

df = pd.DataFrame(data)

print(df)
```

Output:

```
    Study_Hours  Pass
0             1     0
1             2     0
2             2     0
3             3     0
4             3     1
5             4     0
6             4     1
7             5     1
8             6     1
9             7     1
10            8     1
11            9     1
```

Step 3 — Separate X and y

```
X = df[["Study_Hours"]]
y = df["Pass"]
```

Here:

```
X = Study_Hours
y = Pass
```

Step 4 — Create Logistic Regression model

```
model = LogisticRegression()
```

Step 5 — Train the model

```
model.fit(X, y)
```

Now the model learns the relationship between study hours and passing.

Step 6 — Make a prediction

Let's predict whether a student who studies **5 hours** will pass.

```
new_student = [[5]]

prediction = model.predict(new_student)

print("Prediction:", prediction[0])
```

You might get:

```
Prediction: 1
```

Meaning:

```
1 = Pass
0 = Fail
```

Step 7 — Get probability

This is one of the important differences between Logistic and Linear Regression.

```
probability = model.predict_proba(new_student)

print(probability)
```

For example:

```
[[0.20 0.80]]
```

This means:

```
20% → Fail
80% → Pass
```

You can print it more clearly:

```
probability = model.predict_proba(new_student)[0]

print("Fail Probability:", probability[0])
print("Pass Probability:", probability[1])
```

Step 8 — Check predictions for all students

```
df["Prediction"] = model.predict(X)

print(df)
```

Now you have:

```
    Study_Hours  Pass  Prediction
0             1     0           0
1             2     0           0
2             2     0           0
3             3     0           0
4             3     1           0
5             4     0           1
6             4     1           1
7             5     1           1
8             6     1           1
9             7     1           1
10            8     1           1
11            9     1           1
```

Step 9 — Plot the data

First, plot the actual Pass/Fail values:

```
plt.scatter(X, y)
```

```
plt.xlabel("Study Hours")
plt.ylabel("Pass / Fail")
plt.title("Study Hours vs Pass/Fail")

plt.show()
```

Step 10 — Plot the Logistic Regression curve

For a proper logistic curve, create many X values:

```
import numpy as np

X_range = np.linspace(1, 9, 100).reshape(-1, 1)

probability = model.predict_proba(X_range)[:, 1]

plt.scatter(X, y, label="Actual Data")

plt.plot(X_range, probability, label="Logistic Curve")

plt.xlabel("Study Hours")
plt.ylabel("Probability of Passing")
plt.title("Logistic Regression")

plt.legend()
plt.show()
```

You will see an **S-shaped curve**.

Step 11 — Find the model coefficient

```
print("Coefficient:", model.coef_[0][0])
print("Intercept:", model.intercept_[0])
```

The logistic regression equation starts with:

$z=b0+b1X$

and then the sigmoid function converts z into probability:

$P=1+e−z1$

Complete code

Once you understand each step, you can use this shorter complete version:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LogisticRegression

# 1. Create DataFrame
data = {
    "Study_Hours": [1, 2, 2, 3, 3, 4, 4, 5, 6, 7, 8,
9],
    "Pass":        [0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1]
}

df = pd.DataFrame(data)

# 2. X and y
X = df[["Study_Hours"]]
y = df["Pass"]

# 3. Create model
model = LogisticRegression()

# 4. Train model
model.fit(X, y)

# 5. Predict
new_student = [[5]]

prediction = model.predict(new_student)
probability = model.predict_proba(new_student)[0]

print("Prediction:", prediction[0])
print("Fail Probability:", probability[0])
print("Pass Probability:", probability[1])

# 6. Coefficient and intercept
```

```
print("Coefficient:", model.coef_[0][0])
print("Intercept:", model.intercept_[0])

# 7. Plot Logistic Curve
X_range = np.linspace(1, 9, 100).reshape(-1, 1)
probability_curve = model.predict_proba(X_range)[:, 1]

plt.scatter(X, y, label="Actual Data")
plt.plot(X_range, probability_curve, label="Logistic
Curve")

plt.xlabel("Study Hours")
plt.ylabel("Probability of Passing")
plt.title("Logistic Regression")

plt.legend()
plt.show()
```

Remember this flow □

```
DataFrame
    ↓
X = Study Hours
y = Pass/Fail
    ↓
LogisticRegression()
    ↓
model.fit(X, y)
    ↓
model.predict()
    ↓
model.predict_proba()
    ↓
Probability
    ↓
Class 0 or 1
```